Yang Wu

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Education

B.S. in Mathematics & B.A. in Music Composition 2023 – 2027
Soochow University — GPA: 3.9/4.0 Taipei City, Taiwan

Neuropsychiatric Disorder Summer Associate School 2024
National Taiwan University & International Brain Research Organization Taipei City, Taiwan

Cognitive Neuroscience Summer School 2024
National Central University Taoyuan City, Taiwan

Research Experience

Flexible cognitive map building, without synaptic plasticity 2025/7 – Present
University of Oxford; Mentors: James Whittington & Joseph Warren

- Identified a neural algorithm that organises sequence memory dynamically, thereby generalising a loop-based sequential predictive task. After learning from just 8 loop lengths, the model reaches > 0.99 accuracy to 8 other never-seen-before loop lengths.
- Neural algorithm: When the underlying loop length is unknown, the Prefrontal Cortex (PFC) model flexibly re-partitions its neural space for every new sensory input, providing as many memory-storing subspaces as the current loop needs. Thereby correctly infer the loop length every time.
- Abstract under review for COSYNE 2027, first-author preprint in preparation.

Hippocampal theta sweeps on an emerged cognitive map 2025/7 – Present
New York University; Mentors: Marcelo Mattar & Daniel Levenstein (Yale)

- RNN that learns to predict upcoming sensory signals naturally develop a cognitive map. We couple it with a GRU in charge of theta sweep control to test the functionality of theta sweep and goal-directed sweep on such an emerged cognitive map in RNN.
- Episodic simulation through memory recombination: Meta-trained GRU with key-value memory module on a three-phase navigation task (encode motor commands (e.g. move forward for 5 steps, go left for 3 steps) as memories → combining motor commands to form navigation plans → execute plan to reach the goal) to model how past memories could recombined into future plan.

Interpreting planning-like behaviour in naturalistic foraging RL agents 2025/7 – 2025/12
Harvard University; Mentor: Ryan Badman in Kanaka Rajan's lab

- Extended the decoders analysis on model-free reinforcement learning agents from PPO-GRU to a PQN-LSTM architecture, showing its neural activity predicts where the agent will move over the next 10-30 steps beyond what its position and recent movement predict.
- Methods: Per-neuron GLMs (NeMoS); linear decoders of past and future position; sparse autoencoders on hidden states, matched to behavioural data.
- Deep RL Needs Deep Behavior Analysis (NeurIPS 2025; acknowledged in the paper)

Decomposing inter-regional neural dynamics during cued reaching in mice 2024/7 – 2025/7
Academia Sinica; Mentor: Yu-Wei Wu

- Applied Current-Based Decomposition (CURBD) to model inter-regional communication from neural activity recorded in M1, M2, and S1.
- CURBD reproduced our M1/M2/S1 recordings. Yet the inter-regional currents inferred cannot be confirmed, due to our dataset only had one cued reaching scenario; a useful negative result on CURBD's data requirements.

EMG gesture detection & LLM music generation for remote-rehabilitation 2024/2 – 2025/3
Academia Sinica; Mentor: Yu-Te Wang

- Classified 8 gestures using CNN from 3-channel EMG; Developed pipeline of real-time LLM music generation based on the rehabilitation gestures detected from EMG.
- A new type of EEG Electrodes: Collected human EMG/EEG/SSVEPs data; Helped build watermill-like EEG electrodes that better reach human scalp through various hairstyles.
- Publications & Presentations in IEEE journal, SfN, IEEE workshop, National Conference

Publications & Presentations

[**5**] **A neural mechanism for dynamic structuring of memories during schema inference**
Under review for COSYNE 2027, in preparation for preprint 2026
Yang Wu, Joseph Warren, James Whittington

[4] 3D printed watermill-like semi-dry electrodes for BCI applications
IEEE Transactions on Neural Systems and Rehabilitation Engineering 2026
Chi-Ming Chung, Ching-Hung Tsai, Yu-Lin Chu, Chih-Hao Hsu, Jui-Bang Lu, Yun-Chieh Hsu, Yu-Jung Su, **Yang Wu**, Cian-Fong Hung, Yu-Te Wang

[**3**] **HCI Technologies in rehabilitation: enhancing physical therapy with EMG, VR and LLM-Generated music**
Poster, Society for Neuroscience Annual Meeting 2024
Yang Wu*, Yu-Jung Su, Jui-Bang Lu, Yu-Te Wang*, Tong-Hou Cheong

[**2**] **RehabVerse: Mixed Reality Solution for Telerehabilitation**
Poster, National Biotechnology Research Park Pitch Day 2024
Yang Wu*, Yu-Jung Su, Jui-Bang Lu, Yi-Huan Chen, Yu-Te Wang*

[1] BrainPrint: Innovative Head-Mounted EEG Technology for Secure Personal Identification
Poster, IEEE Brain Discovery & Neurotechnology Workshop 2024
Yu-Lin Chu, Ching-Chih Tsao, Chi-Ming Chung, Cian-Fong Hung, Yao-Yu Lee, Jui-Bang Lu, **Yang Wu**, Yi-Huan Chen, Yu-Te Wang*

Awards, Grants & Extracurricular

Ministry of Education Study Abroad Programme for Future Scholars in Humanities and Social Sciences (US$50,000) 2026
Computational and Systems Neuroscience (COSYNE) Conference Undergraduate Travel Grant (US$1,500) 2025
Chu-Chin Hsu Scholar (Top 2 in class) 2025
Dean's List (Top 3 in class) × 3 2024 – 2025
Finalist, National Biomedical Engineering Award (BrainPrint) 2024
Distinction Award, International ICT Innovation Services Award (BrainPrint) 2024
National Science and Technology Council Neuroscience Camp 2024
Academia Sinica Statistical Science Camp 2024

Courses & Skills

Selected Courses (grades on 100-point scale): **Computational Neuroscience (Grad Level) 95**, **Statistical Mechanics (Grad Level) 87**, **Advanced Calculus (Real Analysis) 99**, Linear Algebra 96, Programming in Life Science 97, Life Science (Physiology) 95.

Skills: **Data collection (EEG, EMG, SSVEPs)**; Python; Bayesian inference; Deep learning (PyTorch, RNNs, Transformers); Reinforcement learning (model-free and model-based, PPO, GRPO); Statistical modeling (GLMs, sparse autoencoders, linear decoders); Tools (Git, Blender, MATLAB, 3D printing).

References

James Whittington; University of Oxford; james.whittington@psy.ox.ac.uk

Marcelo Mattar; New York University; mm13100@nyu.edu

Daniel Levenstein; Yale University; daniel.levenstein@yale.edu

Ryan Badman; Harvard University; ryan_badman@hms.harvard.edu

Yu-Wei Wu; Academia Sinica; wuyuwei@gate.sinica.edu.tw

Yu-Te Wang; Academia Sinica; yutewang@citi.sinica.edu.tw

Music

Milonga — Latin-American piece for Cello and Piano. vimeo.com/1036267350

Zhang Beihai said goodbye to his father — Solo Piano. vimeo.com/1036267291

Weathering With You Movie Trailer (Re-score). vimeo.com/1036267308